



Aravindakshan R R 250701066-2025-CSE ▾

A2

Time left 0:59:20

Question 1 | Correct Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence: [-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n;
5      scanf("%d", &n);
6      int a[n];
7      int dp[n];
8      for (int i = 0; i < n; i++) {
9          scanf("%d", &a[i]);
10         dp[i] = 1;
11     }
12     int max = 1;
13     for (int i = 1; i < n; i++) {
14         for (int j = 0; j < i; j++) {
15             if (a[j] <= a[i]) {
16                 if (dp[j] + 1 > dp[i]) {
17                     dp[i] = dp[j] + 1;
18                 }
19             }
20         }
21         if (dp[i] > max) {
22             max = dp[i];
23         }
24     }
25     printf("%d", max);
26     return 0;
27 }
```

Check

	Input	Expected	Got	
✓	9 -1 3 4 5 2 2 2 2 3	6	6	✓
✓	7 1 2 2 4 5 7 6	6	6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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